

Lists - Ordered List, Unordered List and Definition List

List-related tags are meant for marking up a list of items. HTML supports three types of lists: ordered list, unordered list and definition list.

i. Unordered List `<ul>...</ul>` and List Item `<li>...</li>`

Function: An unordered list is shown with a bullet in front of each item. The `<ul>...</ul>` contains an unordered list. Each of items in the list is enclosed in `<li>...</li>`, as follows:

```
<ul>
  <li>list-item-1</li>
  <li>list-item-2</li>
  .....
</ul>
```

Example:

```
<p>An OMO web designer must master:</p>
<ul>
  <li>Hypertext Markup Language (HTML)</li>
  <li>Cascading Style Sheet (CSS)</li>
  <li>HyperText Transfer Protocol (HTTP)</li>
  <li>Apache HTTP Server</li>
</ul>
```

Output of the example: An OMO web designer must master:

- Hypertext Markup Language (HTML)
- Cascading Style Sheet (CSS)
- HyperText Transfer Protocol (HTTP)
- Apache HTTP Server

ii. Ordered List `<ol>...</ol>` and List Item `<li>...</li>`

Items in an ordered list are numbered automatically by the browser. The container tag `<ol>...</ol>` contains an ordered list. Each item of the list is contained inside a `<li>...</li>` container tag. The syntax is similar to the unordered list.

Example:

```

<p>An OMO web designer must master:</p>
<ol>
  <li>Hypertext Markup Language (HTML)</li>
  <li>Cascading Style Sheet (CSS)</li>
  <li>HyperText Transfer Protocol (HTTP)</li>
  <li>Apache HTTP Server</li>
</ol>

```

iii. Definition List <dl>...</dl>, Definition Term <dt>...</dt> and Definition Detail <dd>...</dd>

Function: <dl>...</dl> tag contains a Definition List. Each of <dt>...</dt> and <dd>...</dd> pair contains a Definition Term and the Definition Detail, as follows:

```

<dl>
  <dt>term-1</dt>
  <dd>definition-for-term-1</dd>
  <dt>term-2</dt>
  <dd>definition-for-term-2</dd>
  .....
</dl>

```

Example:

```

<p>These are some of the commonly-encountered web protocols:</p>
<dl>
  <dt>HTTP</dt><dd>HyperText Transfer Protocol</dd>
  <dt>FTP</dt><dd>File Transfer Protocol</dd>
  <dt>SMTP</dt><dd>Simple Mail Transfer Protocol</dd>
  <dt>POP</dt><dd>Post Office Protocol</dd>
  <dt>PPP</dt><dd>Point-to-point Protocol</dd>
  <dt>SLIP</dt><dd>Serial Line Internet Protocol</dd>
</dl>

```

Output of the example:

HTTP: HyperText Transfer Protocol

FTP: File Transfer Protocol

SMTP: Simple Mail Transfer Protocol

POP: Post Office Protocol

PPP: Point-to-point Protocol

SLIP: Serial Line Internet Protocol

The "unordered list" and "ordered list" are used in most HTML documents. But I don't find many web pages using the "definition list".

iv. Nested Lists

Example 1:

```
<p>The topics covered are:</p>
<ul>
  <li>HyperText Markup Language (HTML)
    <ul>
      <li>Based on SGML</li>
      <li>Used to create web pages</li>
      <li>Maintained by W3C</li>
    </ul>
  </li>
  <li>Cascading Style Sheet (CSS)
    <ul>
      <li>Used to define presentation style for web pages</li>
      <li>Also maintained by W3C</li>
    </ul>
  </li>
</ul>
```

Example 2:

```
<ol>
  <li>XML: Extensible Markup Language
    <ol>
      <li>Based on SGML</li>
      <li>Maintained by W3C</li>
    </ol>
  </li>
  <li>DOM: Document Object Model</li>
  <li>SAX: Simple API for XML</li>
</ol>
```

Tables

Table-related tags are meant for tabulating data. (Older HTML documents tend to use `<table>` for formatting the document to divide the document into columns/sections, which should be avoided. Use style sheets for formatting instead.)

The basic unit of a table is a cell. Cells are grouped into rows. Rows are grouped to form the table. This corresponds well to the "row-centric" approach in the display.

The tags used by table are:

- `<table>...</table>`: contains the entire table.
- `<tr>...</tr>`: contains a row.
- `<th>...</th>` and `<td>...</td>`: contain a header cell and a data (detail) cell respectively.
- `<caption>...</caption>`: specifies a caption.
- `<thead>...</thead>`, `<tbody>...</tbody>`, and `<tfoot>...</tfoot>`: for table header, body, and footer.
- `<colgroup>...</colgroup>` and `<col>...</col>`: for applying styles to column group and column respectively.

For Example:

```
<table>
  <caption>Price List</caption>
  <tr>
    <th>Fruit</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Apple</td>
    <td>$0.50</td>
  </tr>
  <tr>
    <td>Orange</td>
    <td>$0.65</td>
  </tr>
</table>
```

Output:

Fruit	Price
Apple	\$0.50
Orange	\$0.65

Table `<table>...</table>`

Function: Set up a table, consisting of rows of cells. Three optional presentation attributes, `border="n"`, `cellspacing="n"`, and `cellpadding="n"`, are often used in older HTML pages but now deprecated. The now-preferred approach is to use CSS.

Table Row `<tr>...</tr>`

Function: Set up a row inside a table, consisting of cells.

Table Header Cell `<th>...</th>`, Table Data Cell `<td>...</td>`

Function: Set up each individual cell of a row. `<th>...</th>` defines a header cell (usually displayed in bold with center alignment) and `<td>...</td>` defines a body cell.

Attributes "rowspan" and "colspan"

On a `<td>` cell, we can use the attribute `rowspan="numOfRows"` or `colspan="numOfColumns"` to span the cell to occupy multiple rows or columns.

Example 1:

```
<table>
  <tr>
    <td>11111</td>
    <td>22222</td>
    <td>33333</td>
  </tr>
  <tr>
    <td>44444</td>
    <td rowspan="2">55555</td>
    <td>66666</td>
  </tr>
  <tr>
    <td>77777</td>
```

```
        <td>88888</td>
    </tr>
</table>
```

Example 2:

```
<table>
  <tr>
    <td colspan="2" rowspan="2">11111</td>
    <td>22222</td>
  </tr>
  <tr>
    <td>33333</td>
  </tr>
  <tr>
    <td>44444</td>
    <td>55555</td>
    <td>66666</td>
  </tr>
</table>
```

Table Header <thead>...</thead>, Table Body <tbody>...</tbody> and Table Footer <tfoot>...</tfoot>

Function: Used to define a header, body and footer sections for a table. Browsers may duplicate the header or footer when breaking the table across multiple pages.

Column Group <colgroup>...</colgroup> and Column <col />

Function: <colgroup>...</colgroup> can be used to group a set of columns so that style can be applied to all. The attribute `span="numOfColumns"` specifies the number of columns.

Anchor and Link <a>...

A hypertext-link allows users to:

1. Navigate to a different document.

2. Navigate to an "Anchor Point" (bookmark).
3. Request other Internet resources (such as e-mail).

Videos and Audio Files

The easiest way to add video or sound is to include the `<embed>` tag. You can also include a `<noembed>` tag for older browsers.

Attributes:

- **Align:** Determines object alignment (center, left, right).
- **Autostart:** Boolean to start automatically.
- **Loop:** Specifies continuous play.
- **Hidden:** Hide or show the object.
- **Src:** URL of the object.

Embed Video Example:

```
<video controls width="250">  
  <source src="/media/cc0-videos/flower.mp4"  
    type="video/mp4">  
</video>
```

Embed Audio Example:

```
<audio controls src="/media/cc0-audio/t-rex-roar.mp3">  
</audio>
```

HTML Forms

HTML forms are required to collect data from visitors (e.g., login, registration).

Form Element

Wraps all other elements.

```
<form action="index.html" method="post">  
</form>
```

Attributes:

1. **action:** The URL of the program that processes the information.
2. **method:** The HTTP method (POST or GET).

Input Element

Used to make text fields.

```
<input type="text" id="name" name="student_name">
```

Attributes:

- **type:** Kind of input (text, email, password, etc.).
- **id:** Used to associate labels or for CSS/JS.
- **name:** Necessary for server-side processing.
- **placeholder:** Short hint displayed in the field.

Textarea Element

Used for multiple lines of text.

```
<textarea id="bio" name="student_bio"></textarea>
```

Button Element

Used to submit forms. Types include `submit`, `reset`, and `button`.

```
<button type="submit">Submit</button>
```

Label Element

Identifies form controls. The `for` attribute matches the `id` of the input.

```
<label for="name">Name</label>
```


Select Menus

Dropdown list of options.

```
<select id="courses" name="student_courses">
  <optgroup label="Engineering">
    <option value="cs">Computer Science</option>
  </optgroup>
</select>
```

Radio Buttons

Used for picking one option from a small group.

```
<input type="radio" id="under_16" value="under_16" name="user_age">
```

Checkboxes

Used for selecting multiple options.

```
<input type="checkbox" id="engineering" value="eng" name="user_interest">
```

Difference Between Checkbox and Radio Button

1. Checkboxes can exist alone; radio buttons must be in groups.
2. Checkboxes are optional; radio buttons usually require one choice.